

XACLE CHALLENGE 2026: THE FIRST X-TO-AUDIO ALIGNMENT CHALLENGE

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ABSTRACT

We present the task description and results of ICASSP 2026 SP Grand Challenge GC-12, “x-to-audio alignment,” which aims to predict the semantic alignment between general audio and text for evaluating text-to-audio generation. The goal is to develop an automatic evaluation method that correlates highly with human evaluations. In the final results, 16 out of 18 teams outperformed the baseline, with the top-5 teams achieving improvements of nearly 0.3 in Spearman’s rank correlation coefficient.

Index Terms— x-to-audio generation, text-to-audio generation, objective evaluation, subjective evaluation, audio–language model

1. INTRODUCTION

Generating general audio from various inputs, such as text and video (x-to-audio generation; XTA), has been actively studied [1, 2, 3, 4, 5]. In XTA, both subjective and objective evaluations of how well the output matches the input are important [6, 7]. For instance, in the text-to-audio generation (TTA), methods have been proposed to evaluate the alignment between audio and text objectively [8, 9]. However, these methods often have a low correlation with human subjective evaluations [10, 11].

Figure 1 illustrates an overview of this task. The goal of this task is to develop a model that automatically predicts the alignment score between audio and text, explicitly targeting high correlation with human subjective scores. Such an evaluation framework is essential for the long-term goal of faithfully generating audio that follows human instructions. Moreover, it will be helpful for understanding human audio perception. For these reasons, this task constitutes an important and timely initiative for advancing XTA research.

2. TASK SETUP

2.1. Dataset

Table 1 shows the statistics of the dataset. The training, validation, and test datasets consist of the following:

- **Audio–text pairs.** All texts are written in English. All audio samples are provided in mono 16-bit 16 kHz RIFF WAV format.
- **Listener-level semantic alignment scores.** The dataset includes subjective semantic-alignment scores between audio

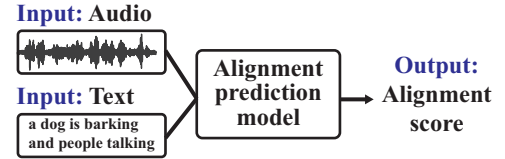


Fig. 1. Overview of task.

and text assigned by individual listeners. The scores are measured on an 11-point scale from 0 (“does not match at all”) to 10 (“matched exactly”). Each audio–text pair was evaluated by native English speakers, with four listeners per pair for the training and validation datasets and eight listeners per pair for the test datasets.

- **Average scores per audio–text pair.** In addition to listener-level scores, this dataset provides the average score for each audio–text pair, computed across all listeners.
- **Listener IDs who gave the semantic-alignment scores.** The dataset also includes anonymized listener IDs associated with each semantic-alignment score.

The audio samples consist of natural and synthesized sounds. The natural sounds used in the training and validation datasets are selected from AudioCaps [12]. The natural sounds included in the test dataset are newly collected from YouTube, and their text descriptions were obtained via a crowdsourcing service¹. The synthesized sounds are generated by eight TTA models, AudioLDM [1], AudioLDM 2 [2], Tango [13], Tango 2 [14], AudioGen [15], TangoFlux [16], Stable Audio², and Make-An-Audio 2 [17], using the text descriptions corresponding to the natural sounds as input.

For each audio–text pair, a semantic-alignment score is obtained by presenting the audio and the corresponding text description to a listener and asking the question, “How would you rate the relevance of the audio to the description above?”

During the challenge, for the test dataset, only the audio–text pairs and the average scores were released to the participants.

2.2. Evaluation metrics

We use metrics that demonstrate correlations and score differences from human evaluation scores. Submissions will be evaluated on the correlation coefficient and score error between predicted and average-semantic-alignment scores. Specifically, the metrics include the linear correlation coefficient (LCC) [18], Spearman’s rank correlation coefficient (SRCC) [19], Kendall’s rank correlation coefficient

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¹<https://www.prolific.com/>

²<https://stableaudio.com/>

Table 1. Statistics of train, validation, and test datasets.

	Training	Validation	Test
#Evaluations	30,000	12,000	12,000
#Audio-text pairs	7,500	3,000	3,000
Audio durations [s]	75,000	30,000	30,000
#total Listeners	2,323	668	1,336

(KTAU) [20], and mean squared error (MSE) referring to the Voice-MOS Challenge [21]. Let $\mathbf{y}$ and $\hat{\mathbf{y}}$ be averaged human scores and automatically predicted scores, respectively, the evaluation metrics are calculated as follows.

$$\text{SRCC} = 1 - \frac{6 \sum d_i^2}{N(N^2 - 1)}, \quad (1)$$

$$d_n = \text{rank}(y_n) - \text{rank}(\hat{y}_n), \quad (2)$$

where N and $\text{rank}(\cdot)$ denote the number of samples and sorting by rank. If there are ties, the average rank is assigned to each of the tied values.

$$\text{LCC} = \frac{\sum_{n=1}^N (y_n - m_y)(\hat{y}_n - m_{\hat{y}})}{\sqrt{\sum_{n=1}^N (y_n - m_y)^2} \sqrt{\sum_{n=1}^N (\hat{y}_n - m_{\hat{y}})^2}}, \quad (3)$$

where m_y and $m_{\hat{y}}$ denote the mean of $\mathbf{y}$ and $\hat{\mathbf{y}}$, respectively.

$$\text{KTAU} = \frac{N_c - N_d}{\sqrt{(N_c + N_d + N_{tx})(N_c + N_d + N_{ty})}}, \quad (4)$$

where N_c , N_d , N_{tx} , and N_{ty} denote the number of concordant pairs where the ranks of the predicted and average human scores, discordant pairs, tied pairs for the x-axis variable, and tied pairs for the y-axis variable.

$$\text{MSE} = \frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n)^2, \quad (5)$$

The final ranking is determined based on the SRCC metric. If multiple teams are tied, the standings will be determined using the LCC, KTAU, and MSE metrics.

2.3. Task rules

External datasets and pre-trained models. Participants can use external data in addition to the official training set, if they declare its use to the organizers and receive permission. Pre-trained models are subject to the same rule: only the models declared to the organizers and approved by them may be used. If participants use pre-trained models, they must also declare to the organizers what data were used to train those models. External data and pre-trained models are limited to those that are open-access. It is prohibited to collect new audio samples, texts, and scores for model training.

Model training. There will be no specific restrictions on the model size and inference time.

2.4. Baseline system

We have provided a baseline model following the RELATE [22]. It consists of audio encoder (pre-trained BYOL-A [23]), text encoder (pre-trained RoBERTa [24]), and a long short-term memory (LSTM) [25]-based score predictor. The audio and text are input to the audio and text encoders, respectively. The feature vectors extracted by

Table 2. Results of top-ranked and baseline systems.

Team name	Test				Validation
	SRCC $\uparrow$	KTAU $\uparrow$	LCC $\uparrow$	MSE $\downarrow$	SRCC $\uparrow$
Sun.NPU	0.6402	0.6873	0.4612	3.0111	0.6680
Chunarkar.NTHU	0.6382	0.6851	0.4596	2.8256	0.6645
Shiota.TMU	0.6327	0.6426	0.4564	9.5999	0.6780
Huang.WHU	0.6264	0.6695	0.4497	2.8369	0.7082
Guan.HEU	0.6143	0.6770	0.4403	2.8044	0.6455
Baseline	0.3345	0.3420	0.229	4.8113	0.3844

each encoder are concatenated along the feature dimension and fed into the score predictor to estimate the score. The model is publicly available³.

3. CHALLENGE RESULTS

Overall results. A total of 18 teams and 48 systems were submitted. Table 2 shows the evaluation results of the top five teams. The top-performing teams were able to develop systems that significantly outperform the baseline model. Although not reported in this paper, 16 out of the 18 teams constructed systems that outperform the baseline model, demonstrating substantial progress in the task of predicting subjective semantic alignment between audio and text. We computed the LCC between the SRCC scores on the validation and test datasets across all systems. As a result, a high correlation of 0.928 was observed, indicating that the relative trends among systems are consistent between the validation and test datasets. This also supports that no substantial domain shift occurs between the validation and test datasets.

Trend in models and external datasets. CLAP-derived models (e.g., M2D-CLAP [26]) were used as both audio and text encoders in 34 of the 48 systems, confirming their effectiveness for this task. Focusing only on the text encoder, we observed that DeBERTa was used in multiple systems, following CLAP models.

Focusing on the score predictors, top-performing systems adopted Transformer-based or cross-attention-based architectures combined with multi-layer perceptrons (MLPs). Overall, Transformer combined with MLPs was the most commonly used configuration. In contrast, the second-ranked team used a support vector regression (SVR) for the score predictor. In addition, we observed a weak correlation (LCC = 0.360) between the number of model parameters and SRCC on the test dataset. This suggests that, under the current task setting, large-scale models are not necessarily required to predict audio-text semantic alignment.

Ten systems utilized external datasets, and teams ranked in the top five tended to use them. They were primarily paired environmental sound-text datasets such as AudioCaps for pre-training.

4. CONCLUSION

We presented the task description and the results of the ICASSP 2026 SP Grand Challenge GC-12. This task aims to develop an automatic evaluation method for semantic alignment between audio and text that correlates highly with human evaluations. Most of the submitted systems outperformed the baseline model, and the top-performing teams achieved improvements of nearly 0.3 in SRCC over the baseline. We hope that the results of this challenge and the submitted technical reports will contribute to the advancement of subsequent research.

³https://github.com/XACLE-Challenge/the_first_XACLE_challenge_baseline_model

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